

Muhammad Blal

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EDUCATION

Iowa State University

Bachelor of Science in Computer Science (GPA: 3.50/4.0)

Expected Graduation: **December 2027**

Ames, IA

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, JavaScript, TypeScript, SQL, HTML, CSS, Dart, Swift

AI/ML & Data: Scikit-learn, Pandas, NumPy, Matplotlib, Ollama, GitHub Copilot, Claude Code, ChatGPT Codex

Frameworks: Angular, React.js, Next.js, Node/Express.js, Flask, Spring Boot, Quarkus, FastAPI

Cloud/DevOps: Azure, AWS, EKS, Kubernetes, Terraform, Docker, Helm, CI/CD, Git, Postman, REST APIs

Databases: MongoDB, PostgreSQL, Neo4j, Elasticsearch

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Principal Financial Group

Des Moines, IA

- Developed **Terraform** modules, **ArgoCD** pipelines, and **TypeScript/JavaScript** automation for AWS EKS infrastructure on the Infrastructure Services Containers team, supporting containerized enterprise workloads serving **18,000+ employees**.
- Converted **25+ Kyverno ClusterPolicies** to **CEL** and integrated **40+ CloudWatch/Dynatrace alerts** with ServiceNow and xMatters, reducing manual EKS configuration and incident routing effort by **30%** across **5+** engineering teams.

Software Engineer Intern

May 2025 – Aug 2025

J.B. Hunt Transport Services, Inc.

Lowell, AR

- Engineered a net-new Angular/TypeScript feature for J.B. Hunt 360 personnel and driver panels, improving usability for **35,000+ daily users**.
- Enhanced Java microservices with Spring Boot, Quarkus, MongoDB, Elasticsearch, and Kubernetes CI/CD canary deployments, contributing to projected savings of **\$250,000** over 5 years.

Software Engineering Simulation (Forage Program)

Jun 2025 – Jul 2025

JPMorgan Chase

Remote

- Designed and deployed a **Kafka-based** transaction processing system in **Spring Boot**, improving transaction validation speed by **30%** and **ensuring 100%** data integrity in an **H2 SQL database**.
- Integrated an **external Incentives API** to dynamically apply rewards, optimizing real-time balance updates and enhancing transaction accuracy.

Web Application Developer

Aug 2024 – Mar 2025

XO Foods (Chester's)

Waterloo, IA

- Built a full-stack restaurant website using React.js, Node.js, and Express, increasing online orders by in a **35%**.
- Optimized responsive UI components, SEO, and Google Analytics tracking, improving mobile engagement by **50%** and monthly traffic by **20%**.

TikTok Shop Affiliate & E-Commerce Content Creator

Sep 2023 – Present

TikTok

Remote

- Generated over **\$50,000 in TikTok Shop sales** through short-form product marketing content, leveraging audience trends and product research to drive engagement and conversions.
- Analyzed content performance, conversion metrics, and product trends to optimize product selection and marketing strategies across affiliate campaigns.

PROJECTS

Bitcoin Price Predictor | Python, Scikit-learn, Pandas, Matplotlib, yFinance

Jul 2026 – Aug 2026

- Built a machine learning model with lag features and technical indicators, improving price prediction accuracy by **15%** compared to baseline methods.
- Automated data via **yFinance API** and built iterative forecasts projecting Bitcoin trends into **2026**, with outputs saved as CSV/PNG visualizations.

EasyCal | React, Node.js, Express, MongoDB, FastAPI, Python, Ollama

Feb 2025 – May 2025

- Built a full-stack nutrition and recipe planning web app used by **Iowa State University students**, featuring dynamic food search, recipe tracking, restaurant listings, and full **CRUD operations**.
- Integrated an AI-assisted recipe generator using **Ollama LLM** and **FastAPI**, boosting user engagement by over **40%** and reducing manual content entry by **50%**.

LEADERSHIP

Computer Science and Software Engineering Club (CSE) | Member

September 2024 – Present

Pakistan Student Association | Member

September 2024 – Present